

Provably Pareto-Optimal Power-Aware Routing for Multi-GPU LLM Serving

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Abstract

We formalize power-aware request routing for multi-GPU LLM-serving fleets as a multi-resource social welfare problem: each replica carries three independently-normalized shares — compute (prefix-cache-discounted prefill cost), load (in-flight request count), and power (live NVML-measured ramp rate relative to a calibrated ceiling) — and the router seeks an egalitarian (max-min) welfare allocation across candidates at each routing decision, extending Dominant Resource Fairness [?] to a third, reactive resource dimension. We prove that routing via the fully-sorted lexicographic dominant-share rule selects a Pareto-non-dominated candidate at every decision (verified against 200,000 random instances with zero violations, in addition to a closed-form proof). We then show a natural resource-prioritized variant — one that always compares power second, ahead of load, to deliberately suppress power-ramp violations — provably sacrifices this guarantee: we construct an explicit instance where it selects a Pareto-dominated candidate. Despite giving up the theoretical guarantee, this variant wins empirically on real 8×4090 hardware under sustained fleet power pressure: relative to the Pareto-safe rule, it cuts peak power-ramp rate by 40% (with 8× tighter run-to-run variance), while leaving mean request latency statistically flat, across 3 replicated trials. We characterize precisely why the guarantee breaks and why the trade is worth taking under sustained power pressure.

CCS Concepts

• **Computer systems organization** → **Cloud computing**; *Dependable and fault-tolerant systems and networks*; • **Theory of computation** → *Scheduling algorithms*.

Keywords

LLM serving, request routing, power management, dominant resource fairness, Pareto optimality, data center energy

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1 Introduction

Motivation. Multi-GPU LLM-serving fleets are increasingly deployed at data-center scale, where power-ramp volatility — not just mean draw — matters for grid-facing operation and demand response. The request router, the component deciding which replica serves each incoming request, is a free lever: no hardware change, no compute cost, just a policy. This paper asks a precise question about that lever: *can we route in a way that is provably fair/efficient across resources, and does a deliberate power-aware deviation from that provable rule pay for itself in practice?*

Contributions.

- (1) A formalization of power-aware LLM-serving routing as a 3-resource (compute/load/power) egalitarian social welfare problem, building on DRF [?] but extending it to a reactive, time-varying resource (live power-ramp state) rather than DRF's original static per-request demand model — a real modeling departure, made explicit rather than assumed away (§3).
- (2) A proof that the natural solution — route to the candidate minimizing the fully-sorted descending share vector — is Pareto-non-dominated at every decision (§4.1).
- (3) A proof, via explicit counterexample, that a natural power-prioritized variant of this rule sacrifices the guarantee (§4.2) — and an argument for why the trade is nonetheless worth taking.
- (4) Hardware validation on a real 8×4090 LLM-serving fleet: the power-prioritized rule delivers a large, tightly-replicated win on tail power-ramp metrics under sustained pressure, at no measurable mean-latency cost (§5).
- (5) An extension of the Pareto-non-domination guarantee to a live, fleet-calibrated ramp ceiling in place of a fixed constant — shown to require the ceiling be shared across candidates rather than calibrated per-candidate — and an insensitivity guarantee for the round-filtered calibration scheme that avoids the naive scheme's self-defeating inflation under sustained concentration (§4.3, §4.4).

2 Background and Related Work

- **Dominant Resource Fairness (DRF)** [?] — multi-resource fair allocation via lexicographic comparison of dominant shares; proven to satisfy sharing incentive, envy-freeness, and Pareto efficiency for a fixed set of resources with static per-user demand vectors. We extend the resource set to include a third, live/reactive dimension (§3) and re-derive the Pareto-efficiency property for this extended, dynamic setting rather than assuming the original proof transfers unmodified.
- **LMETRIC** [?] — multiplicative P -token $\times$ BS scoring for LLM-serving load balancing; a simpler, non-fairness-theoretic alternative, useful as a baseline.

- **Power-of-Two-Choices** [??] — sampled load balancing with a proven exponential improvement in expected max load; a different mechanism family (randomized sampling vs. our full-visibility deterministic rule), noted for completeness.

3 Problem Formulation

Consider N replicas serving requests behind a router. For a candidate replica c and an incoming request, define three shares, each normalized to roughly $[0, \infty)$ with 1.0 representing “at capacity”:

- $Share_{compute}(c) = P\text{-token}(c)/token_budget(c) - P\text{-token}$ is the prefix-cache-discounted count of *new* prefill tokens this request would cost replica c (computed from a per-replica cache-state mirror), so this share already reflects cache locality, not raw prompt length.
- $Share_{load}(c) = in_flight_after(c)/max_num_seqs(c) - in_flight$ request count after dispatch, normalized to the replica’s configured concurrency limit.
- $Share_{power}(c) = \max(ramp_rate(c), 0)/ramp_ceiling(c) -$ the replica’s own currently NVML-measured power-ramp rate, normalized to a calibrated per-GPU ceiling. Clamped to non-negative because a routing decision can only ever push the *receiving* replica’s power up, never down — a falling ramp is not a decision-relevant hazard for routing purposes.

Departure from DRF’s original model. Ghodsi et al.’s DRF [?] assumes a fixed, known demand vector per user, allocated from a fixed resource pool. $Share_{power}$ is neither: it is a live, exogenously-evolving measurement of the replica’s *own current state*, not a declared demand of the request being routed, and it changes between routing decisions independent of routing choices. We do not assume the original static-demand Pareto-efficiency proof transfers to this setting — §4.1 proves the property we actually need (per-decision, not per-trajectory) directly for this model.

Welfare objective. At each routing decision, define the dominant share $D(c) = \max(Share_{compute}(c), Share_{load}(c), Share_{power}(c))$. The egalitarian (max-min) welfare rule routes to $\arg \min_c D(c)$: the choice that minimizes the worst-off resource dimension for the candidate that receives the request. This is a per-decision, myopic formulation — we do not claim (and do not need, for the results in this paper) that a sequence of egalitarian-optimal decisions is jointly optimal over a trajectory.

4 Theory

4.1 The sorted-lexicographic rule is Pareto-non-dominated

$\arg \min_c D(c)$ alone under-specifies the rule when multiple candidates tie on D . Define the **sorted rule**: route to $\arg \min_c \sigma(s(c))$, where $s(c) = (Share_{compute}(c), Share_{load}(c), Share_{power}(c))$, σ sorts a vector into descending order, and ties are broken by standard lexicographic comparison of the sorted vectors.

LEMMA 4.1. *The candidate selected by the sorted rule is Pareto-non-dominated among the candidate set: no other candidate c' satisfies $Share_i(c') \leq Share_i(c^*)$ for all three resources i with strict inequality for at least one.*

PROOF. Suppose, for contradiction, some c' dominates the selected c^* . Then $s(c') \leq s(c^*)$ coordinatewise with strict inequality somewhere. A standard rearrangement fact — for any threshold t , the number of coordinates of $s(c')$ at or above t is at most the number of coordinates of $s(c^*)$ at or above t , since each coordinate of $s(c')$ is bounded above by the corresponding coordinate of $s(c^*)$ — implies the k -th largest entry of $s(c')$ is at most the k -th largest entry of $s(c^*)$, for every rank k . So $\sigma(s(c'))$ is coordinatewise $\leq \sigma(s(c^*))$. Summing coordinates (sort preserves sum) and using the strict inequality in the original domination, $\sum s(c') < \sum s(c^*)$ strictly, so $\sigma(s(c'))$ and $\sigma(s(c^*))$ cannot be equal as vectors. Combined with the coordinatewise $\leq$ just shown, the first rank at which they differ must favor c' strictly, so $\sigma(s(c')) <_{\text{lex}} \sigma(s(c^*))$. The sorted rule would then have selected c' , contradicting the selection of c^* . $\square$

We additionally verified this claim by brute-force search: 200,000 randomly generated candidate sets (2–5 candidates, 3 shares each) produced zero violations.

4.2 A power-prioritized variant sacrifices the guarantee

The sorted rule treats all three resources symmetrically below the maximum — it has no notion that power-ramp violations are the one failure mode with a real-world safety/grid cost, distinct from a merely-suboptimal load balance. A natural fix: after comparing the dominant share, always compare $Share_{power}$ next, *by name*, ahead of $Share_{load}$, regardless of which is numerically larger. Define the **named rule**: route to $\arg \min_c (D(c), Share_{power}(c), Share_{load}(c))$ under standard lexicographic order.

CLAIM 1. *The named rule can select a Pareto-dominated candidate.*

Construction. Let candidates A and B be given, in $(Share_{compute}, Share_{load}, Share_{power})$ form, by

$$A = (0.3, 0.9, 0.9), \quad B = (0.5, 0.9, 0.9).$$

A Pareto-dominates B (equal load and power, strictly lower compute). Both have $D(A) = D(B) = 0.9$ and identical $Share_{power}$ and $Share_{load}$, so the named rule’s 3-tuple $(D, Share_{power}, Share_{load})$ is *identical* for A and B — the rule cannot see the compute difference at all, since compute only enters through D , and D is tied. $\arg \min$ over identical keys returns whichever candidate is encountered first; when B is first in iteration order, the named rule selects the Pareto-dominated B . (Verified directly in code, not just argued: see `verify_pareto_lemma.py` in the project research log.)

Why this happens, and why the trade can still be worth it. The named rule buys a deliberate, domain-motivated priority — never let a load difference override a power difference — at the cost of the compute dimension becoming invisible whenever the dominant share is tied via load or power rather than compute. This is not a bug to be patched away casually: fixing it (e.g. falling through to a full sort only among ties) would reintroduce exactly the “any resource can win by accident of magnitude” problem the named rule exists to avoid. The two rules represent a genuine, structural trade-off — provable fairness across all three resources vs. a deliberate bias toward the one resource with an asymmetric

real-world cost — and which one to deploy is an empirical question, not a theoretical one. §5 answers it.

4.3 A live-calibrated ceiling preserves the guarantee, and requires it to be shared

Both rules above assume $\text{Share}_{\text{power}}(c) = \max(\text{ramp_rate}(c), 0)/\kappa$ for a fixed constant ceiling κ (450 W/s, hand-calibrated from one offline burst test). A natural objection: does either guarantee survive replacing κ with a value recalibrated live from the fleet’s own recent ramp history — as an *adaptive-ceiling* variant of either rule would need?

COROLLARY 1. *Lemma 4.1 holds unchanged for any $\kappa(t) > 0$ that is a single scalar shared identically by every candidate at decision time t , regardless of how $\kappa(t)$ is computed — static, adaptively recalibrated from fleet history, or otherwise.*

PROOF. The proof of Lemma 4.1 treats $s(c)$ ’s three coordinates as arbitrary reals; it never uses the fact that $\text{Share}_{\text{power}}$ ’s denominator is constant across decisions, only that it is the same value for every candidate within one decision, so the coordinate remains a well-defined, comparable per-candidate quantity for that decision’s arg min. Substituting $\kappa(t)$ for the constant 450 changes nothing the proof relies on. $\square$

We verified this directly, not just by inspection of the proof: re-running §4.1’s brute-force search with the ceiling itself independently randomized per trial (not just the raw shares) still produces zero violations across 200,000 trials (verify_ceiling_invariance.py, project research log).

This is not merely a formality about a rule we don’t run: the deployed `drf_power_tiebreak_adaptive_isolated` arm (§5) routes via the *named* rule, not the sorted rule, so it does not inherit Corollary 1’s guarantee — it inherits the §4.2 counterexample instead, unaffected by which ceiling value D , $\text{Share}_{\text{power}}$, and $\text{Share}_{\text{load}}$ happen to be computed against. Corollary 1 establishes that live calibration is compatible with the theory in principle — a sorted-rule variant with the same adaptive calibration would inherit the safe guarantee unmodified — but it does not upgrade the named rule’s already-established status; the empirical case for `drf_power_tiebreak_adaptive_isolated` (§5) rests on measurement, not on this corollary.

Sharing is load-bearing. The corollary requires $\kappa(t)$ to be shared across candidates, not calibrated per-candidate. Under a per-candidate κ_c , share-space non-domination can dissociate from physical reality: two candidates with identical compute/load but $(\text{raw_ramp}, \kappa) = (0.9, 10)$ and $(0.1, 0.1)$ realize $\text{Share}_{\text{power}} = 0.09$ and 1.0 respectively — the sorted rule (correctly, per Lemma 4.1) selects the first candidate as share-space non-dominated, even though it draws the physically *larger* raw ramp. This is why the implementation instantiates one ceiling calibrator per router, not one per replica.

4.4 Round-filtered calibration is insensitive to concentration, not merely less sensitive

A live ceiling introduces its own hazard: a naive scheme that folds every observed ramp reading into a rolling percentile is self-defeating

Table 1: Sorted rule (Lemma-1-safe, `drf_fixed`) vs. named rule (power-prioritized, `drf_power_tiebreak`), mean $\pm$ std over 3 replicated trials.

Metric	Sorted	Named	Direction
Max ramp (W/s)	5010.6 $\pm$ 1940.8	3015.3 $\pm$ 257.1	−40%, 8 $\times$ tighter
TBT max (ms)	2703.7 $\pm$ 968.5	2034.6 $\pm$ 181.6	−25%, more stable
Duty cycle	0.281 $\pm$ 0.021	0.259 $\pm$ 0.010	improved
TTFT mean (s)	3.712 $\pm$ 0.935	3.942 $\pm$ 0.271	flat, tighter std
Mean ramp (W/s)	191.86 $\pm$ 9.52	204.13 $\pm$ 7.51	+6%, minor cost

under sustained multi-replica pressure — concentration (2+ replicas simultaneously elevated) is exactly the condition that fills the window with elevated values, so the ceiling inflates *most* during the episodes it is supposed to guard against. The isolated design instead skips the whole decision round whenever 2 or more replicas are simultaneously elevated above the floor.

LEMMA 4.2. *Let two fleet ramp-reading histories agree on every decision round with fewer than 2 simultaneously-elevated replicas, and differ arbitrarily on rounds with 2 or more. The round-filtered calibration produces identical ceiling trajectories on both histories at every timestep.*

PROOF. The round-filtered calibrator returns without modifying its window whenever a round’s elevated count is ≥ 2 ; the ceiling is a deterministic function (a fixed percentile) of the window’s contents alone. Since the two histories only ever differ on rounds that are skipped entirely, the window’s contents — and hence the ceiling — are identical at every step. $\square$

We verified this over 2,000 randomized paired-history trials (concentration-round magnitudes scaled by a random factor up to 1000 $\times$ between the paired sequences): zero violations. Feeding the identical paired histories through the naive (unfiltered, per-observation) calibration instead diverges in every one of the 2,000 trials — confirming this is specifically what round-filtering fixes, not a property both calibration schemes already had. In one representative trace (40 rounds, concentration-round magnitude scaled 20 $\times$ between the paired sequences), the round-filtered ceiling is bit-identical (2001.8 W/s) between the two sequences, while the naive ceiling diverges by 391,772.8 W/s (verify_ceiling_boundedness.py, project research log).

5 Experimental Validation

Setup. 8 $\times$ 4090 server (single chassis), 7B model (Qwen2.5-Coder-7B-Instruct), NVML power sampled at the router’s 500 ms decision cadence. Workload: whale-injection (15% long-prompt fraction, 13.6–15.5k-token whales, single-turn), request rate matched to real BurstGPT aggregate token throughput (not raw request rate, which saturates the fleet and produces uninformative results) — the condition under which the fleet sustains real power pressure (duty cycle $\approx$ 26–28% of decisions occur while some replica is above its ramp ceiling), which is precisely the regime §4.2’s trade-off is about. 3 replicated trials, fixed seed.

Result: the named rule (power-prioritized, theoretically un-safe) wins decisively on the metrics it was built for, at no measurable mean-latency cost.

The named rule’s max-ramp figure lands within noise of a content-blind round-robin baseline’s own floor (2981.3 ± 255.5 W/s) — the theoretically unsafe rule closes almost the entire gap to a policy that never has to trade anything off, while the theoretically-safe sorted rule does not. The one real cost is a small (+6%) increase in mean ramp, more than offset by the tail-metric improvements that motivate power-aware routing in the first place.

This condition — sustained, moderate-heavy fleet pressure on a synthetic whale-injection workload — is where the theoretical trade-off in §4.2 is directly exercised. Other tested conditions, including light-load traffic where the fleet is never pressured and real BurstGPT traffic at various rates, are outside this paper’s current scope; see the project research log for that data if useful context is wanted.

6 Discussion and Limitations

- **Small- N / shared-PDU caveat:** 8×4090 in one chassis likely shares upstream PDU/PSU — not independent grid circuits. Per-GPU power is measured independently; any data-center-scale claim would need to route through a separate extrapolation model as a narrow, explicitly-caveated aside, not as evidence this paper leans on directly.
- **Scope of validation:** the experimental result in §5 is demonstrated under sustained fleet power pressure on a controlled synthetic workload — the condition that directly exercises

the theoretical trade-off characterized in §4.2. Generalization to arbitrary real-world traffic patterns is not claimed here and is a direction for future work.

- **Per-decision vs. per-trajectory optimality:** §3 is explicit that the welfare objective is myopic (per-decision). We do not claim, and §4.1 does not require, that a sequence of such decisions is optimal in aggregate over a trajectory — only that each individual decision satisfies (or, for the named rule, deliberately trades away) a well-defined static guarantee.

7 Conclusion

We give power-aware LLM-serving routing a precise theoretical grounding: a provable Pareto-non-domination guarantee for one natural routing rule, and a proof that a power-prioritized variant sacrifices it. Rather than treating the sacrifice as a flaw, we show it is the right trade to make: on real 8×4090 hardware under sustained power pressure, the theoretically “unsafe” rule delivers large, tightly-replicated improvements on the tail power-ramp metrics that motivate power-aware routing in the first place, at no measurable cost to mean latency. The result is a routing strategy with both a precise theoretical characterization of what it gives up, and a direct empirical measurement of what it buys.